interacting with something else. They materialize in a place, with a calculable probability, when colliding with something else. The ‘quantum leaps’ from one orbit to another are the only means they have of being ‘real’: an electron is a set of jumps from one interaction to another. When nothing disturbs it, it is not in any precise place. It is not in a ‘place’ at all.

It’s as if God had not designed reality with a line that was heavily scored, but just dotted it with a faint outline.

In quantum mechanics no object has a definite position, except when colliding headlong with something else. In order to describe it in mid-flight, between one interaction and another, we use an abstract mathematical formula which has no existence in real space, only in abstract mathematical space. But there’s worse to come: these interactive leaps with which each object passes from one place to another do not occur in a predictable way but largely at random. It is not possible to predict where an electron will reappear, but only to calculate the *probability* that it will pop up here or there. The question of probability goes to the heart of physics, where everything had seemed to be regulated by firm laws which were universal and irrevocable.

Does it seem absurd? It also seemed absurd to Einstein. On the one hand he proposed Heisenberg for the Nobel Prize, recognizing that he had understood something fundamental about the world; whilst on the other he didn’t miss any occasion to grumble that this did not make much sense.

The young lions of the Copenhagen group were dismayed: how was it possible that *Einstein* should think this? Their spiritual father, the man who had shown the courage to think the unthinkable, now retreated and was afraid of this new leap into the unknown which he himself had triggered. The same Einstein who had shown that time is not universal and that space is curved was now saying that the world cannot be *this* strange.

Patiently, Bohr explained the new ideas to Einstein. Einstein objected. He devised mental experiments to show that the new ideas were contradictory: ‘Imagine a box filled with light, from which we allow a single photon to escape for an instant …’ So begins one of his famous examples, the mental experiment of the ‘box of light’. In the end Bohr always managed to find an answer with which to rebut these objections. For years, their dialogue continued by way of lectures, letters, articles… During the course of the exchange both great men needed to backtrack, to change their thinking. Einstein had to admit that there was actually no contradiction within the new ideas. Bohr had to recognize that things were not as simple and clear as he’d